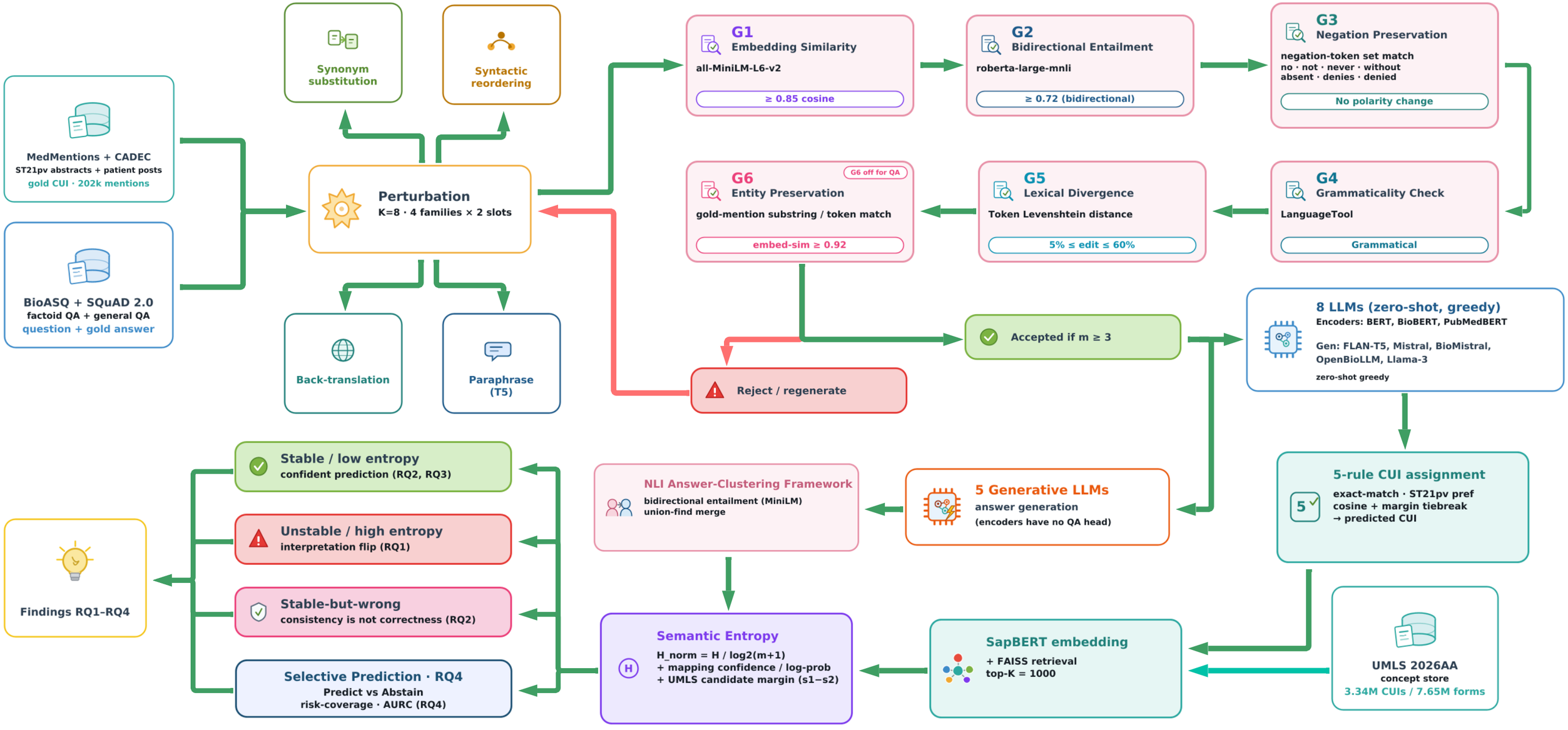

MedMentions + CADEC
ST21pv abstracts + patient posts
gold CUI · 202k mentions
BioASQ + SQuAD 2.0
factoid QA + general QA
question + gold answer
Synonym substitution
Syntactic reordering
Perturbation
K=8 · 4 families × 2 slots
Back-translation
Paraphrase (T5)
G1
Embedding Similarity
all-MiniLM-L6-v2
≥ 0.85 cosine
G2
Bidirectional Entailment
roberta-large-mnli
≥ 0.72 (bidirectional)
G3
Negation Preservation
negation-token set match
no · not · never · without
absent · denies · denied
No polarity change
G4
Grammaticality Check
LanguageTool
Grammatical
G5
Lexical Divergence
Token Levenshtein distance
5% ≤ edit ≤ 60%
G6
G6 off for QA
Entity Preservation
gold-mention substring / token match
embed-sim ≥ 0.92
Reject / regenerate
Accepted if m ≥ 3
8 LLMs (zero-shot, greedy)
Encoders: BERT, BioBERT, PubMedBERT
Gen: FLAN-T5, Mistral, BioMistral, OpenBioLLM, Llama-3
zero-shot greedy
5-rule CUI assignment
exact-match · ST21pv pref
cosine + margin tiebreak
→ predicted CUI
5 Generative LLMs
answer generation
(encoders have no QA head)
NLI Answer-Clustering Framework
bidirectional entailment (MiniLM)
union-find merge
Semantic Entropy
H_norm = H / log2(m+1)
+ mapping confidence / log-prob
+ UMLS candidate margin (s1−s2)
SapBERT embedding
+ FAISS retrieval
top-K = 1000
UMLS 2026AA
concept store
3.34M CUIs / 7.65M forms
Stable / low entropy
confident prediction (RQ2, RQ3)
Unstable / high entropy
interpretation flip (RQ1)
Stable-but-wrong
consistency is not correctness (RQ2)
Selective Prediction · RQ4
Predict vs Abstain
risk-coverage · AURC (RQ4)
Findings RQ1–RQ4